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Observations on the thermal conductance of Adélie (*Pygoscelis adeliae*) and Humboldt (*Spheniscus humboldti*) penguins

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Abstract Analyses of cooling rates in one Adélie and one Humboldt penguin yielded calculated thermal conductance values of 0.1040 and $0.1672 \text{ W}(\text{kg}^\circ\text{C})^{-1}$, respectively. We review the methods used to calculate penguin surface area, an important component in calculating conductance values, and suggest that, in comparative studies of thermal balance, the use of body mass is a better estimator of body size than surface area. Using previously published data on penguin species, we found a significant model to predict thermal conductance from body mass according to: $\log C = \log 0.1083 - 0.474 \log M$, where C is minimal specific thermal conductance in $\text{W}(\text{kg}^\circ\text{C})^{-1}$ and M is body mass in kilograms.

Introduction

Maintenance of body temperature necessitates that endotherms balance heat production with heat loss. Both these parameters can vary widely, but of the two, variation in heat loss, which is regulated via changes in thermal conductivity, involves the least expenditure of energy, and is thus expected to be the principal regulatory mechanism. Thermal conductivity is usually defined as heat flux from an animal's core to its environment (Richards 1973) and it can be determined experimentally using three methods: (1) via determination of the standard metabolic rate (Davenport 1992); (2) by using cooling rates of corpses (Herreid and Kessel 1967); and (3) by measuring the difference in temperature between the core of the study animal and the surface of its insulating medium e.g. fur, feathers

(Veghte and Herreid 1965). To estimate thermal conductivity in birds and mammals, Herreid and Kessel (1967) developed an empirical model using the cooling rates of corpses that relates thermal conductivity exponentially to animal body mass. For birds, excluding penguins, these authors proposed that conductivity relates to body mass by: $\log C = 0.662 - 0.536 \log M$, where M is the mass in grams and conductivity (C) is measured in $\text{Cal}(\text{g h}^\circ\text{C})^{-1}$.

As part of a larger study on the feeding ecology and thermoregulation of penguins (e.g. Wilson et al. 1992; Wilson et al. 1995), Adélie penguins in Antarctica, and Humboldt penguins in Chile were equipped with stomach temperature loggers. During the experiments one equipped, free-living Adélie penguin and one captive Humboldt penguin accidentally died. After death, the stomach temperature loggers continued recording as the bird body temperature cooled until the bird stomach temperature was equivalent to that of the environment. This allowed us to obtain data on cooling rates from which we were able to determine thermal conductance. Our major aim was to evaluate differences in thermal conductance with respect to the different thermal conditions under which Adélie and Humboldt penguins live, and to compare our results with published thermal conductance data from other penguin species. However, in revising the literature, we found that comparison between thermal conductance data of different species is difficult due to the different methodologies used to determine body surface area, a critical parameter. Consequently, this paper reviews the literature on thermal conductance and includes our opportunistic data.

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Materials and methods

The work was conducted on an Adélie penguin at Ardley Island $62^\circ 13' \text{ S}$, $58^\circ 55' \text{ W}$), Antarctica on 29 December 1991 and on a Humboldt penguin at the Instituto de Investigaciones

Oceanológicas, Antofagasta, Chile, on a bird previously captured at Pan de Azúcar Island (26°09' S, 70°41' W), northern Chile, on 14 February 1994. After death, the stomach temperature loggers continued recording, and we analysed the rate of heat loss from both birds.

Two types of stomach temperature logger were used: the EATL (Einkanaler Automatischer Temperatur Logger, Elくてk Elektronik, Eching, Germany, recording interval 16 s, 32 Kbytes maximum memory, used on the Adélie penguin) and the SICUP (Single Channel Unit Processor; Driesen & Kern, Bad Bramstedt, Germany, recording interval 8 s, 128 Kbytes maximum memory, used on the Humboldt penguin). Both units were powered by 6-V lithium batteries, used quartz crystals for timing and recorded temperature data between 20 and 45°C with 8-bit resolution. Relative and absolute accuracies were better than 0.1 and 0.8°C, respectively, for both units. The electronics of both units were sealed inside titanium tubes with O-ring fitted lids tested as watertight to a pressure of 50 bar. The EATL weighed 80 g (100 × 23 mm diameter) while the SICUP weighed 32.5 g (93 × 19 mm diameter). More details of the EATL and SICUP are given by Wilson et al. (1995).

A few hours after the EATL was administered to an Adélie penguin, the bird died. The bird (mass 4.0 kg) was then placed on an abandoned wind-protected nest in an active breeding colony. Weather conditions were cloudy and overcast, the air temperature was -0.3°C and the wind speed was 5.6 m s⁻¹ (data obtained from the meteorological office of the Chilean Antarctic base, Presidente Eduardo Frei Montalva, a distance of 2 km from study site). The Humboldt penguin died during respirometry experiments. A stomach temperature logger had been administered to the bird in order to measure core temperature simultaneously with determination of oxygen consumption with changing ambient temperature. The carcass (mass 3.85 kg) was maintained in a laboratory (no wind, no insolation) under a constant ambient temperature of 26°C and then placed on a plastic operating table in a position as closely approximating that of a prone, sleeping bird as possible. A thermocouple connected to a chart recorder (YSI, model 81A) was used to continuously measure ambient temperature.

The times for the birds to cool from 32 to 26°C and 34 to 29°C in the Adélie and Humboldt carcasses, respectively, were used to determine the cooling constant and conductance using the method of Morrison and Tietz (1957). According to these authors, the calculated cooling constant may be expressed in units of thermal conductance, in J s⁻¹, by employing the specific heat capacity of the tissue, taken to be 3.47 J(g°C)⁻¹ (Schmidt-Nielsen 1990), and the body mass. The data were subsequently expressed in terms of unit body surface or body mass.

To estimate the surface area of the penguins we used the equation of Meeh: $A = k \cdot M^{0.667}$ (quoted in Drent and Stonehouse 1971) relating bird body mass (kg) to surface area (m²), where k is a constant that describes the geometric shape of a given body (e.g. $k = 0.06$ for a cube and $k = 0.048$ for a sphere, Schmidt-Nielsen 1984), as well as the density of the body. Here we used a $k = 0.1$ as suggested by Drent and Stonehouse (1971).

It was not possible to correct heat loss values for loss due to evaporation since this is governed by complex laws that are markedly different in living and dead specimens (Richards 1973).

Results and discussion

At the instant the penguins died, the temperature gradient between the body and the ambient temperature began to drop by $1.4 \times 10^{-5} \text{ °C s}^{-1}$ (standard error of the estimate = 3.2×10^{-8}) and $2.1 \times 10^{-5} \text{ °C s}^{-1}$ (standard error of the estimate = 4.2×10^{-8}) in the Adélie and Humboldt penguins respectively. Using the equation given by Morrison and Tietz (1957), we calculated

thermal conductances of 1.651 and 2.617 W(m²°C)⁻¹ or 0.1040 and 0.1672 W(kg °C)⁻¹ for the Adélie and Humboldt penguins, respectively.

Firstly, we compared our data in W(m²°C)⁻¹ with those data reported in the literature (original units) and found that our calculated thermal conductivities were 74% and 278%, respectively, of those given by Kooyman et al. (1976) and Drent and Stonehouse (1971) for Adélie [$2.23 \text{ W(m}^2\text{°C)}^{-1}$] and Humboldt penguins [$0.94 \text{ W(m}^2\text{°C)}^{-1}$]. However, Kooyman et al. (1976) reported that the thermal conductivity of Adélie penguin skin (1.8 cm thick including feathers) was $1.75 \text{ W(m}^2\text{°C)}^{-1}$, a value 6% higher than our calculated figure for the entire animal, but 22% lower than the figure these authors reported from living Adélie penguins.

Culik et al. (1991) suggested that differences in measured values of thermal conductivity in penguins may be due to differences in calculations of body surface area. Other authors, such as Barré (1984), Pinshow et al. (1976), and Stahel and Nicol (1982), measured penguin surface areas directly and found k values of 0.087, 0.065 and 0.077, respectively, for little, Adélie and King penguins. Drent and Stonehouse (1971) and Jarman (1973) used $k = 0.1$, as suggested by Benedict (1934; cited in Drent and Stonehouse 1971). Culik et al. (1991) used a k value of 0.12. Kooyman et al. (1976) measured the surface area in one Adélie penguin specimen and calculated an exponent of 0.65, as opposed to 0.667 as suggested originally by Meeh. However, these authors maintained a k value of 0.1. If body surface area cannot be directly measured, what value of k should be used in Meeh's formula, assuming that penguins are isometric? Figure 1 shows the body surface area calculated using various values of k against body mass for a mass range between 0.9 and 27 kg, i.e. from little penguins (*Eudyptula minor*) to emperor penguins (*Aptenodytes forsteri*) (Jarman 1973; Stahel and Nicol 1982). As expected, the

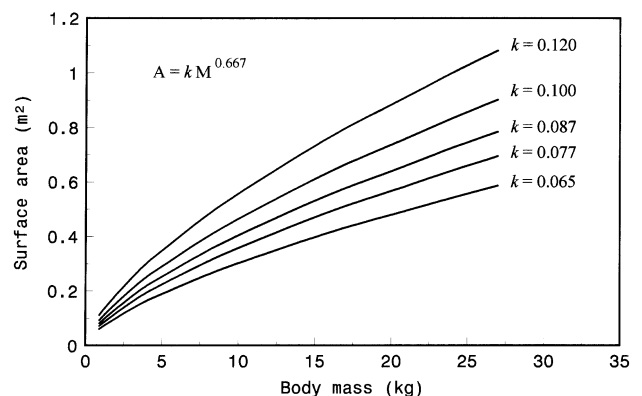


Fig. 1 Effect of variability in the constant k on the calculation of the surface area of penguins. The lines were obtained with Meeh's formula using the different k values reported for penguins and assuming that penguins are isometric in body shape (see text)

discrepancies in the values of *k* produce significant differences in calculated surface area, although all other parameters have been maintained as constant. Two penguin species can, for instance, have similar body masses but vary considerably in proportions (spherical versus elongated). For example, Humboldt and Adélie penguins have a comparable body mass, but differ by approximately 10% with respect to flipper, head and foot size (Lustick 1984). It is clear, therefore, that penguins are not isometric throughout the size range of the family.

A possible alternative is to determine body surface area directly. This parameter has received great attention and several methods for its measurement have been described (cf. Schmidt-Nielsen 1984). However, none of these methods are very reliable and more refined techniques do not seem to improve the accuracy of the measurements (Richards 1973; Schmidt-Nielsen 1984). These two authors provide classical example where values of *k* reported for a single species differed by as much as 50%. For the emperor penguin, the value of *k* used by Barré and Roussel (1986) is 0.077, or 18% higher than the value reported by Pinshow et al. (1976), which was 0.065. However, for animals with very similar body mass (23.40 and 23.37 kg), Barré and Roussel (1986) and Pinshow et al. (1976) reported body surface areas of 0.631 and 0.530 m², respectively. Here, the difference is 16% but the thermal conductance is the same: 1.570 W(m²°C)⁻¹ (see Table 1). The reported

value of Barré and Roussel (1986) was obtained from Barré (1984), who obtained his figure as a mean (0.077 ± 0.009, *n* = 4) from previously published data for emperor penguins (Le Maho et al. 1977; Pinshow et al. 1976; both cited in Barré 1984) and two measurements for king penguins made by the author. Using a *k* value of 0.1 in Meeh's formula, Jarman (1973) estimated a body surface area of 0.899 m² whereas Pinshow et al. (1976) estimated a figure of 0.530 m² (Table 1), derived directly from linear measurements of three standing emperor penguins including the beak and feet. According to Walsberg and King (1978), Meeh's formula estimates the skin surface area beneath the feathers although, with the exception of areas such as the feet and beak, the environmental interface is generally the exterior surface of the plumage. Walsberg and King (1978) determined that skin surface area is, on average, 23% higher than the external feathered surface area. These authors concluded that the use of skin surface area instead of the area of the external plumage surface may produce large errors in heat-transfer analyses. In addition, the body areas available for radiation, convection and conductance of heat are difficult to determine and can, furthermore, be controlled by the animal by changes in posture and tissue irrigation (Davenport 1992). This short analysis shows that the accuracy of the calculated conductance is directly proportional to the accuracy of the estimated body surface area. To summarise, the determination of the surface

Table 1 The thermal conductance of penguins as derived from the literature and expressed using body mass as estimator of body size (*Observed*). The *Predicted* values are from Aschoff's (1981) equations: active phase (α) $\log C = \log 0.946 - 0.484 \log M$; resting (ρ) $\log C = \log 0.947 - 0.583 \log M$. The original units of the equations suggested by Aschoff are ml O₂(g h °C)⁻¹. These units can be converted to W(kg °C)⁻¹ using 1 l O₂ h⁻¹ = 5.579 W (Schmidt-Nielsen

1984); % was calculated using (*Observed/Predicted*) · 100. *Area* is body surface area and *Conductance* is thermal conductance as given by the original sources. Sources are: (1) Stahel and Nicol 1982; (2) Baudinette et al. 1986; (3) Barré and Roussel 1986; (4) This study; (5) Drent and Stonehouse 1971; (6) Kooyman et al. 1976; (7) Chapell and Souza 1988; (8) Barré 1984; (9) Pinshow et al. 1976; (10) Le Maho et al. 1977 and (11) Jarman 1973

Species	Mass kg	Phase	Conductance W(kg °C) ⁻¹			Area m ²	Conductance W(m ² °C) ⁻¹	Source
			Observed	Predicted	%			
<i>Eudyptula minor</i>	0.90	α	0.1567	0.196	80	0.082	1.720	1
<i>E. minor</i>	1.10	α	0.1670	0.178	94	—	—	2
<i>Eudyptes</i> <i>chrysolophus</i>	3.60	α	0.0910	0.100	91	0.181	1.810	3
<i>Spheniscus</i> <i>humboldti</i>	3.85	—	0.1672	0.097	172	0.246	2.617	4
<i>S. humboldti</i>	3.87	ρ	0.0599	0.043	140	0.247	0.940	5
<i>Pygoscelis</i> <i>adeliae</i>	3.97	α	0.1239	0.096	129	0.221	2.230	6
<i>P. adeliae</i>	4.00	—	0.1040	0.095	109	0.252	1.651	4
<i>P. adeliae</i>	4.01	α	0.0712	0.095	75	—	—	7
<i>Aptenodytes</i> <i>patagonicus</i>	8.48	α	0.0460	0.066	69	0.416	0.930	8
<i>A. patagonicus</i>	9.78	α	0.0830	0.062	134	0.458	1.780	8
<i>A. patagonicus</i>	12.45	α	0.0559	0.055	101	0.414	1.680	3
<i>A. forsteri</i>	23.37	α	0.0356	0.041	87	0.530	1.570	9
<i>A. forsteri</i>	23.40	α	0.0423	0.041	104	0.631	1.570	3
<i>A. forsteri</i>	25.00	α	0.0408	0.039	104	0.600	1.698	10
<i>A. forsteri</i>	27.00	α	0.0350	0.038	92	0.899	1.050	11

area is a complex issue whose methodological problems are difficult to eliminate and which make its utilisation in the calculations of thermal conductance questionable (Richards 1973; Schmidt-Nielsen 1984; Walsberg and King 1978).

The alternative is to use the body mass as a measure of animal size. Using body mass avoids errors associated with the determination of surface area and provides useful results for comparative studies (Aschoff 1981; Davenport 1992; Herreid and Kessel 1967; Schmidt-Nielsen 1984). In several studies, the use of body mass has demonstrated its utility in calculation of thermal conductance in arctic birds (e.g. Gabrielsen et al. 1988; Jenssen et al. 1989), aquatic birds (e.g. Visser and Ricklefs 1993) and in studies of the ontogeny of avian thermoregulation (e.g. Montevecchi and Vaughan 1989; Steen and Gabrielsen 1988). In penguins, Baudinette et al. (1986) and Chapell and Souza (1988) also preferred the use of body mass when reporting thermal conductance of little blue and Adélie penguins, respectively.

In order to facilitate inter-specific comparisons, we expressed our results in relation to body mass and compared them with data reported in the literature after conversion (when corresponding) to the same units, taking into account the phase of the diurnal cycle over which the measurements were made (Aschoff and Pohl 1970) (Table 1). All data stem from oxygen consumption measurements made at temperatures below the lower critical temperature of the thermoneutral zone, where thermal conductance is lowest (Aschoff 1981). Data were expressed in relation to the body mass as $W(kg^{\circ}C)^{-1}$, and thus correspond to minimal specific thermal conductance. We found no data on thermal conductance in penguins determined from cooling rates of carcasses.

When plotting penguin minimal specific thermal conductance (Table 1) against body mass on a log/log scale, we obtain a significant regression that is very similar to the model of Aschoff (1981) for non-passerine birds (Fig. 2). The equation obtained is:

$$\log C = \log 0.183 - 0.474 \log M \quad (1)$$

where C is the minimal specific thermal conductance in $W(kg^{\circ}C)^{-1}$ and M is body mass in kilograms ($r = 0.909$; F -ratio = 57.125; $P < 0.001$; $n = 14$). We did not make comparisons with the equation of Herreid and Kessel (1967), because their model underestimates the thermal conductance in birds whose body masses are higher than in their presented mass range (10.3–2.8 kg) (Barré 1984).

The slope of equation 1 (-0.474) compares well to the slope (-0.484) reported by Aschoff (1981), and the difference is not statistically different (test for slope: $P = 0.983$). According to Walsberg and King (1978), the similarity between both slopes may indicate that penguins are not different from non-passerine birds. This contradicts the view of Drent and Stonehouse

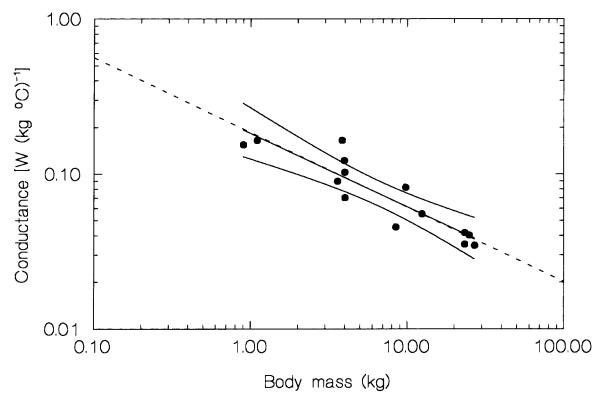


Fig. 2 Specific minimal thermal conductance of penguins. The data of Drent and Stonehouse (1971) are not included because they were obtained during the resting phase of the diurnal cycle (Aschoff 1981). The equation obtained is $\log C = \log 0.183 - 0.474 \log M$ ($r = 0.909$; F -ratio = 57.125; $P < 0.0001$; $n = 14$) and is shown with 95% confidence intervals. The dashed regression line is from Aschoff's (1981) equation for non-passerine birds during the active phase

(1971) who suggested that penguins should be removed from general avian models.

For Adélie penguins, Chapell and Souza (1988) calculated a thermal conductance of $0.0712 W(kg^{\circ}C)^{-1}$. Our calculated thermal conductance [$0.1040 W(kg^{\circ}C)^{-1}$] is 146% of this value, and 84% of the value reported by Kooyman et al. (1976) of $0.1239 W(kg^{\circ}C)^{-1}$ for the entire animal (Table 1). The thermal conductance calculated by us for the Humboldt penguin [$0.1672 W(kg^{\circ}C)^{-1}$] is 279% of the value reported by Drent and Stonehouse (1971) of $0.0599 W(kg^{\circ}C)^{-1}$. These differences can be attributed in part to measurement errors inherent in the different methodologies used. For instance, the heat loss from a dead bird, which is partially dependent on Fourier's laws, cannot be directly likened to that from a living specimen (see Richards 1973). Apart from evaporative heat loss differences, living animals can actively change the conductivity of the skin and feather layer (see Richards 1973; Schmidt-Nielsen 1990). With respect to Humboldt penguins the differences can also be due to the fact that the value reported by Drent and Stonehouse was obtained during the resting phase of the diurnal cycle (Aschoff and Pohl 1970). According to Aschoff (1981) thermal conductance, like many other functions in the organism, depends on circadian phase, and in birds it is about 50% greater when measured during the activity phase of the animal than during the resting phase. For comparison, our calculated thermal conductance for Humboldt penguins is 172% of the predicted value [$0.097 W(kg^{\circ}C)^{-1}$] for the active phase, while the reported value of $0.0599 W(kg^{\circ}C)^{-1}$ obtained by Drent and Stonehouse (1971) is 140% of the expected value for the resting phase (Table 1).

Our results indicate that the thermal conductance of the Humboldt penguin was 62% higher than the thermal conductance of the Adélie penguin. *Pygoscelis adeliae* occurs mainly in continental Antarctica and has

a lower critical temperature of the thermoneutral zone near to -10°C (Chapell and Souza 1988), whereas *Spheniscus humboldti* nests in tropical to temperate regions of Chile and Peru and has a thermoneutral zone ranging between 2 and 30°C (Drent and Stonehouse 1971). Although our data are limited it would seem that the lower thermal conductance of the Adélie compared to the Humboldt penguin reflects an adaptation to colder climates. Similarly, *Eudyptes chrysolophus*, a species with a sub-Antarctic distribution (Williams 1995), has a thermal conductance of $0.091\text{ W}(\text{kg}^{\circ}\text{C})^{-1}$ (Barré and Roussel 1986), a value that is 58% of the thermal conductance of *Eudyptula minor* [$0.1567\text{ W}(\text{kg}^{\circ}\text{C})^{-1}$ Stahel and Nicol 1982], a species restricted to the temperate seas of Southern Australia and New Zealand (Williams 1995). It is generally considered that bird thermal conductance is correlated with the climatic conditions under which they live, with species in colder climates having a lower thermal conductance than those in warmer areas (Calder and King 1974). It is clear that a great many factors (e.g. blood circulation, variable plumage thickness, appropriate thermoregulatory behaviour) may serve to confound the study of thermal conductance of penguins. More work is needed on these factors to enable us to explain satisfactorily the biology and physics of heat loss so as to understand how these tie in with penguin spatial distribution.

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